The Impact of Resampling Techniques on Model Performance and Explainability in Credit Scoring

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Abstract

Credit risk assessment is a crucial task for financial institutions to support lending decisions and managing potential default risks. However, predictive modeling in this domain is often challenged by severe class imbalance, where default cases represent only a small proportion of the data. This study presents an experimental evaluation of several machine learning models, including Logistic Regression, Random Forest, XGBoost, and LightGBM, applied to a real-world credit dataset containing 356,255 observations from the Home Credit Default Risk dataset.To address the imbalance problem, we used three different strategies: cost-sensitive learning, Synthetic Minority Over-sampling Technique (SMOTE), and Random Under-Sampling (RUS), together with hybrid resampling approaches. These techniques were evaluated under multiple sampling ratios (0.2, 0.3, and 0.5) to examine their impact on model performance and interpretability. The experimental results reveal an "accuracy paradox" in baseline models, which achieve high Accuracy scores but significantly low Recall scores for default cases.After applying imbalance mitigation techniques, the models significantly improve minority-class detection. In particular, under-sampling with a ratio of 0.5 provides the most balanced performance in tree-based models, improving recall and F1-score while maintaining stable ROC-AUC values. Furthermore, SHAP-based interpretability analysis shows that external credit scores, credit-to-goods ratio, and annuity-related features consistently dominate model predictions, indicating stable and meaningful risk factors in credit evaluation.

Keywords— Machine Learning, Credit Scoring, Imbalanced Data, Explainable AI (XAI), Resampling Techniques.

1 Introduction

Machine learning plays an essential role in various parts of the finance-banking sector, particularly in credit scoring systems that support banks in assessing customer creditworthiness [1]. Over the past decades, traditional statistical approaches, such as Linear Discriminant Analysis (LDA) and Logistic Regression (LR), have gradually been replaced or enhanced by more complex models, including Random Forest (RF), XGBoost (XGB), and LightGBM [2]. Due to the increased modeling capacity, the accuracy, effectiveness, and cost efficiency of classifying defaulters and non-defaulters have improved significantly [3, 4].

However, one major challenge in credit scoring lies in the highly imbalanced nature of financial datasets, where the proportion of default cases is significantly smaller than the figures for non-default cases [5]. This problem often leads to the favouring of the majority class by models, resulting in poor detection of high-risk customers. Some techniques, such as SMOTE or the adjustment of the threshold in the classifier output, are utilized to address this issue [6]. While these methods can improve predictive performance, their influence on model behavior and interpretability is still not fully understood, especially in high-stakes financial decision-making contexts.

Motivated by the growing demand for responsible and explainable artificial intelligence in banking, this study investigates the use of machine learning models for credit scoring by comparing their predictive performance and examining how the resampling techniques affect model explainability. The studied models take customer demographic and financial attributes as inputs and produce the predicted probability of default as the output. After that, we analyze the impact of data imbalance handling on explanation

methods, such as SHAP and LIME, with the goal of supporting fair and transparent decision-making in credit card and loan approval processes. By demonstrating how preprocessing choices influence not only model predictions but also the interpretation of those predictions, this work contributes to the growing body of research on responsible and explainable artificial intelligence in financial applications.

The remainder of this paper is organized as follows. Section 2 reviews related work on loan default prediction, class imbalance handling, and model explainability. Section 3 describes the machine learning algorithms employed in this study. Section 4 presents the dataset and preprocessing steps. Section 5 discusses the experimental results and findings. Finally, Section 6 concludes the paper.

2 Related work

Credit risk management has undergone a remarkable shift in recent years, moving from traditional statistical methods to more sophisticated machine learning (ML) and deep learning (DL) techniques. However, the increase in predictive accuracy comes with the challenge of interpretability, especially when dealing with financial datasets with high imbalance characteristics.

An empirical study by Mestiri [4] comparing six credit scoring models (including LDA, LR, Decision Trees, Random Forest, SVM and Deep Neural Network) on a consumer loan dataset in Tunisia showed the superiority of modern algorithms. The results showed that the Deep Neural Network (DNN) model achieved the highest performance with an accuracy of 83.6% and an AUC of 0.788, followed by Random Forest (RF). Conversely, traditional statistical models such as LDA yield significantly lower results [4]. This reinforces the trend toward applying nonlinear algorithms to optimize the ability to classify good and bad credit profiles.

Although models like DNN or RF offer superior performance, their "black-box" nature creates a major barrier to trust, especially in sensitive areas like finance. Ribeiro et al. [7] argue that if users do not trust a particular model or prediction, they will not use it. To address this problem, the LIME (Local Interpretable Model-agnostic Explanations) method has been proposed. LIME works by locally approximating a complex model through a simple, understandable model (such as a linear model) around the prediction region of interest. Providing qualitative explanations – for example, indicating which features lead to the rejection of a loan or the classification of an image – helps experts assess the reliability of individual predictions and detect model errors (such as data leakage) that mere accuracy scores cannot reflect [7].

In practical banking applications, credit data frequently suffers from severe class imbalances, where the number of defaulters is much lower than the number of good payers. While much of previous research has focused on improving predictive accuracy on imbalanced data, Chen et al. [5] pointed out a significant research gap: the impact of this imbalance on the stability of explanatory methods such as LIME and SHAP.

Using UK housing mortgage data (2016-2020) along with two other extended credit datasets, the authors established an empirical procedure to measure the stability of explanations as default rates change. The results indicate that as the level of imbalance increases (especially when the bad debt ratio is below 5%), the explanations generated by LIME and SHAP become less stable [5]. Specifically, the rank of the features and their value of importance of the characteristics fluctuate significantly. The study also highlights that LIME tends to be less stable in terms of "external stability" than "internal stability" when faced with extremely imbalanced data.

Although Chen et al. [5] demonstrated that data imbalance significantly reduces the stability of explanations from LIME and SHAP, their research focused primarily on observing the phenomenon through undersampling to create different levels of imbalance. A key question remains whether applying common data balancing techniques (such as SMOTE or oversampling) actually restores the stability of the explanatory model, or introduces further noise into the process. The authors themselves suggested that the potential impact of imbalanced learning techniques on the performance of explanatory methods needs further investigation in future studies.

To address this gap, our study focuses on evaluating the impact of data imbalance mitigation methods on the explanatory power of credit scoring models, specifically through SHAP and LIME. We conducted experiments on a real-world dataset from the "Home Credit Default Risk" competition (Kaggle 2018) [9] and evaluated using multiple models . This is a large dataset with significant imbalances. We used a representative portion of this dataset to test the hypothesis that data imbalance treatment directly impacts the reliability and consistency of model explanations.

3 Theoretical Background

This section provides the mathematical and conceptual foundations for the classification algorithms, imbalance handling techniques, and explainability frameworks employed in this research.

3.1 Classification Algorithms

We employ four distinct classification models, ranging from linear statistical methods to advanced gradient-boosted ensemble techniques.

3.1.1 Logistic Regression (LR)

Logistic Regression is the standard baseline for binary classification in credit risk assessment [1]. It models the probability of the minority class $y = 1$ using the logistic sigmoid function, which maps any real-valued input into the interval $[0, 1]$:

$$\pi(x) = P(y = 1|x) = \frac{1}{1 + e^{-(w^T x+b)}} \tag{1}$$

The model is trained by minimizing the binary cross-entropy (log-loss) function. In credit scoring, the estimated coefficients w represent the log-odds ratio of each financial attribute, allowing for a direct interpretation of how a unit change in a feature affects the risk profile.

3.1.2 Random Forest (RF)

Random Forest is an ensemble meta-estimator that utilizes *Bootstrap Aggregating* (Bagging) and feature randomness to build a forest of uncorrelated decision trees [10]. Each tree is grown on a random subsample of the data, and at each node split, only a random subset of features is considered. The final classification for a query point x is determined by majority voting:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_T(x)\} \tag{2}$$

where T is the number of trees. RF is particularly effective at capturing non-linear interactions between variables and is highly robust to outliers and noise, which are common in credit datasets [6].

3.1.3 eXtreme Gradient Boosting (XGBoost)

XGBoost is a scalable tree boosting system that optimizes a regularized objective function using a sequential additive training process [11]. At each iteration t, it adds a new tree f_t that minimizes the following loss:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} [l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \tag{3}$$

Where g_i and h_i are the first and second-order derivatives of the loss function, respectively. The regularization term $\Omega(f_t) = \gamma T + \frac{1}{2}\lambda ||w||^2$ controls the complexity of the model to prevent overfitting. Its *Sparsity-aware split finding* allows it to handle missing data points efficiently without explicit imputation.

3.1.4 Light Gradient Boosting Machine (LightGBM)

LightGBM improves upon traditional Gradient Boosting Decision Trees (GBDT) by introducing two novel techniques: *Gradient-based One-Side Sampling* (GOSS) and *Exclusive Feature Bundling* (EFB) [12].

- **GOSS:** Keeps all instances with large gradients and performs random sampling on instances with small gradients, maintaining the accuracy of information gain estimation while reducing data volume.
- **EFB:** Bundles mutually exclusive features (features that rarely take non-zero values simultaneously) into a single feature, significantly reducing the dimensionality of the feature space.

Furthermore, LightGBM grows trees leaf-wise (best-first) rather than level-wise, which allows it to achieve lower loss and higher precision on large-scale datasets.

3.2 Handling Class Imbalance

Financial datasets in credit scoring typically exhibit severe class imbalance, which biases models towards the majority class (non-defaulters) and leads to poor recall for the minority class. To mitigate this, we implement a combination of resampling strategies and algorithmic adjustments:

- **Random Under-sampling:** This technique reduces the number of majority class instances to decrease the computational burden and prevent the model from becoming overwhelmed by non-default cases. By removing majority samples, we balance the distribution, though at the risk of losing potentially valuable information.
- **SMOTE (Synthetic Minority Over-sampling Technique):** Instead of simple oversampling by duplication, SMOTE generates synthetic examples by interpolating between a minority sample and its k-nearest neighbors [14]. This technique expands the decision boundary of the minority class, helping the model learn more generalized patterns of default.
- **SMOTE-Tomek (Hybrid Resampling):** This hybrid approach consists of two steps:
  1. **SMOTE:** Applies the oversampling process described above to expand the minority class.
  2. **Tomek Links:** A data cleaning step that identifies and removes "Tomek links"—pairs of very close instances from opposite classes [15]. Removing the majority class instances of these links clears the overlapping regions and sharpens the decision boundary between defaulters and non-defaulters.
- **Cost-Sensitive Learning (Class Weighting):** Complementary to resampling, we modify the loss function by assigning a higher weight W to the minority class. The objective function is defined as:

  $$J(\theta) = -\sum_{i \in \text{majority}} \log(p_i) - W \sum_{j \in \text{minority}} \log(p_j)$$

  This ensures that misclassifying a high-risk customer (False Negative) incurs a significantly higher penalty than a False Positive.

3.3 Explainable AI (XAI): SHAP and LIME

To comply with the "Right to Explanation" in financial regulations [5,8], we employ two post-hoc model-agnostic explanation methods: LIME [7] and SHAP [13].

LIME localizes the interpretation by approximating the black-box model with an interpretable one around specific predictions [7], while SHAP provides a unified framework grounded in cooperative game theory to allocate fair credit to each feature [13].

3.3.1 SHAP (SHapley Additive exPlanations)

SHAP is grounded in cooperative game theory and ensures an additive feature attribution property [13]. The Shapley value for a feature i is calculated as the average marginal contribution across all possible feature subsets S:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \tag{4}$$

This provides a theoretically sound global and local interpretation of the model.

3.3.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME focuses on explaining individual predictions by approximating the "black-box" model locally with an interpretable surrogate model (e.g., a linear regression) [7]. It generates new samples around a specific instance x and weights them based on their proximity to x, ensuring the explanation is locally faithful.

4 Methodology

This section delineates the empirical framework adopted for developing the credit scoring models. The pipeline encompasses data acquisition, rigorous preprocessing, advanced feature engineering, and a robust evaluation strategy designed for imbalanced classification.

4.1 Dataset Description

The empirical analysis is conducted using the *Home Credit Default Risk* dataset [9]. The dataset after merging involves 356,255 total records. A significant challenge identified in the *application_train* corpus is the severe class imbalance, with a majority class (non-defaulters) comprising 91.9% and a minority class (defaulters) representing only 8.1% of the total samples [5]. To rigorously evaluate the models and prevent data leakage, the complete dataset was partitioned into training set and validation set - a hold-out set with a ratio of 80:20. All resampling techniques and model training were exclusively applied to the training set, while the validation set was used for performance benchmarking.

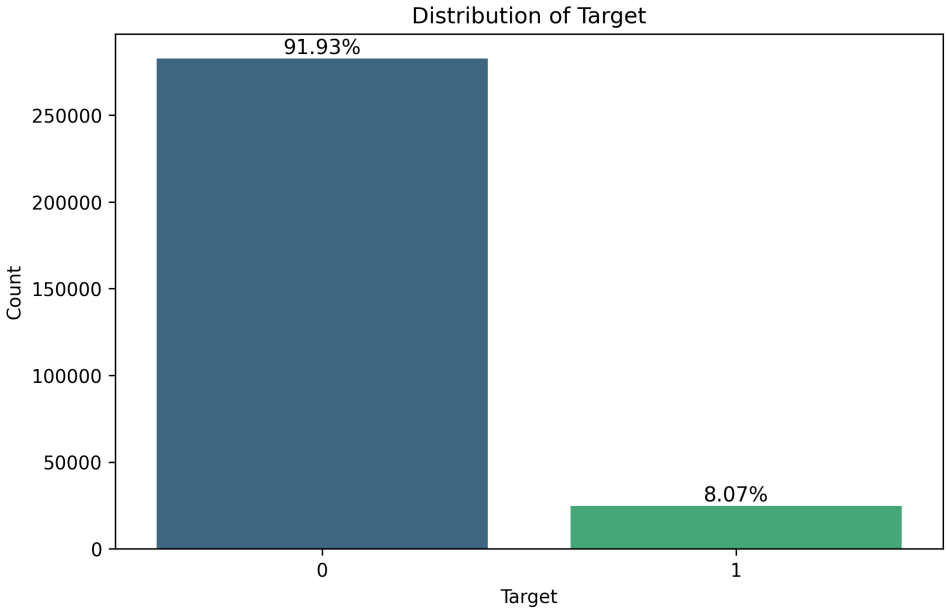

Figure 1: Class Imbalance Ratio

4.2 Data Preprocessing

To ensure the integrity of the predictive modeling, we implemented a multi-stage preprocessing pipeline as executed in our `data_preparation` script:

- **Missing Value Imputation:** For numerical attributes, we employed median imputation to mitigate the influence of skewed distributions [3]. Categorical variables were filled with the most frequent category or a distinct "NA" label.
- **Anomalies Handling:** A notable anomaly was handled in the `DAYS_EMPLOYED` variable, where the erroneous value 365,243 was replaced with `NaN`. Additionally, temporal features were converted from negative days to positive counts (e.g., `AGE`, `EMPLOYMENT_STABILITY`).
- **Logarithmic Transformation:** We applied a log-transformation to stabilize variance:

$$x_{trans} = \ln(1 + x) \tag{5}$$

  This was applied to features such as `LOG_AMT_CREDIT`, `LOG_AMT_ANNUITY`, and `LOG_AMT_GOODS_PRICE`.
- **Feature Encoding:** We utilized Label Encoding for binary features and One-Hot Encoding for multi-categorical variables.

4.3 Feature Engineering and Selection

Based on the data preparation logic, we synthesized several "Domain Features" to better capture applicant risk profiles [1]:

- `CREDIT_BY_INCOME`: The ratio of the total credit to the applicant's income.

- `ANNUITY_BY_INCOME`: The percentage of income consumed by the loan annuity.
- `GOODS_PRICE_BY_INCOME`: The ratio of goods price to income.
- `INCOME_PER_PERSON`: Total income divided by the number of family members.
- `CREDIT_TO_GOODS_RATIO`: The ratio of credit amount to the price of the goods.

To optimize performance, we utilized a **Random Forest-based Feature Selection** method [10]. By leveraging the *Mean Decrease in Impurity* (MDI), we filtered the feature space to retain the top-performing variables.

Table 1: List of variables used in credit score modeling

Name	**Description**
EXT_SOURCE_MEAN	Avg. of external credit scores.
EXT_SOURCE_3	External credit score source 3.
EXT_SOURCE_2	External credit score source 2.
EXT_SOURCE_1	External credit score source 1.
YEARS_EMPLOYED	Total employment duration (years).
ANNUITY_LENGTH	Loan repayment term duration.
CREDIT_TO_GOODS	Ratio: Credit amt / Goods price.
MP_STABILITY	Job tenure relative to client age.
AGE	Client's age in years.
LOG_AMT_ANNUITY	Log-transformed periodic payment.
LOG_AMT_GOODS	Log-transformed price of goods.
DAYS_ID_PUB	Days since last ID change.
DAYS_PHONE_CHG	Days since last phone change.
LOG_AMT_CREDIT	Log-transformed total credit.
ANNUITY_INCOME	Ratio: Annuity / Total income.
DAYS_REGIS	Days since last registration change.
CREDIT_INCOME	Ratio: Credit / Total income.
GOODS_INCOME	Ratio: Goods price / Total income.
REGION_POP	Normalized region population.
INC_PER_PERSON	Disposable income per family member.

Table 2: Descriptive statistics for continuous variables

Variable Name	**Mean**	**St.Dev**	**Min**	**Max**
EXT_SOURCE_MEAN	0.509	0.150	0.000	0.879
EXT_SOURCE_3	0.511	0.195	0.001	0.896
EXT_SOURCE_2	0.514	0.191	0.000	0.855
EXT_SOURCE_1	0.502	0.211	0.015	0.963
YEARS_EMPLOYED	6.536	6.406	0.000	49.074
ANNUITY_LENGTH	21.655	7.846	8.046	48.000
CREDIT_TO_GOODS_RATIO	1.121	0.124	0.222	3.000
EMPLOYMENT_STABILITY	0.180	0.157	0.000	1.000
AGE	43.930	11.956	20.518	69.121
LOG_AMT_ANNUITY	10.053	0.536	7.388	12.461
LOG_AMT_GOODS_PRICE	12.983	0.825	10.610	15.214
CONVERTED_DAYS_ID_PUBLISH	8.204	4.135	0.000	19.718
CONVERTED_DAYS_LAST_PHONE_CHANGE	2.645	2.268	0.000	11.759
LOG_AMT_CREDIT	13.109	0.778	10.714	15.214
ANNUITY_BY_INCOME	0.181	0.095	0.000	1.876
CONVERTED_DAYS_REGISTRATION	13.662	9.651	0.000	67.595
CREDIT_BY_INCOME	3.958	2.690	0.005	33.738
GOODS_PRICE_BY_INCOME	3.535	2.457	0.005	33.738
REGION_POPULATION_RELATIVE	0.021	0.014	0.000	0.073
INCOME_PER_PERSON	84.187	64.915	0.018	515.651

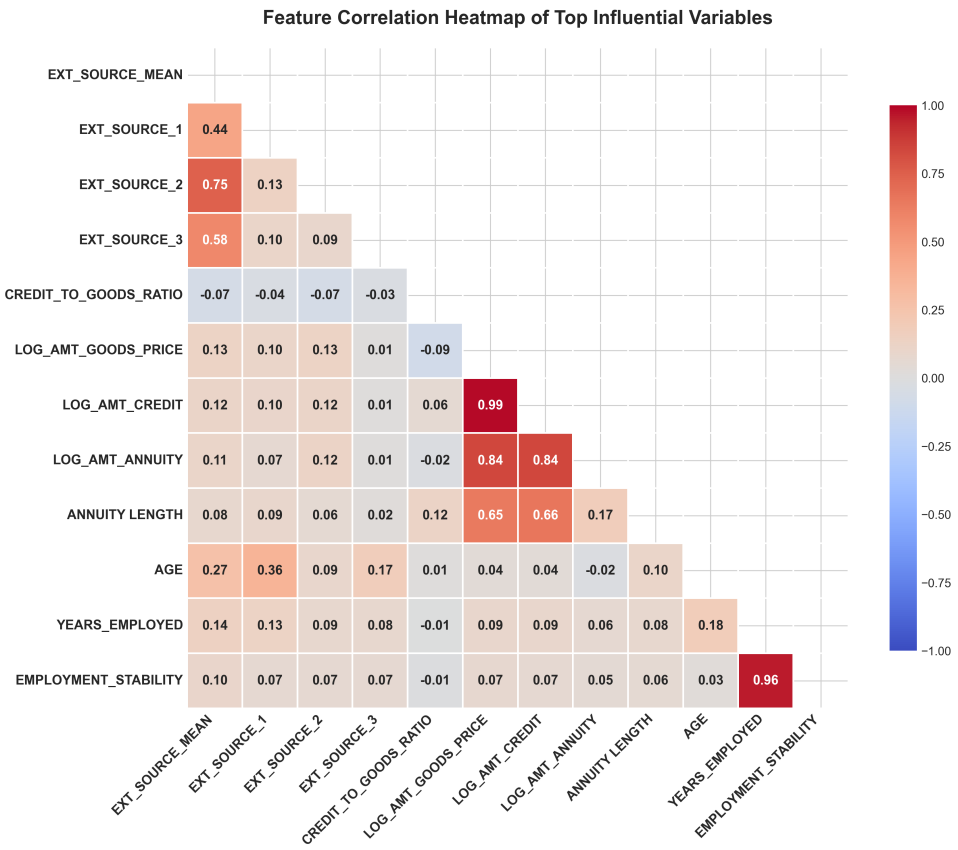

Figure 2: Feature Correlation Heatmap

4.4 Evaluation Metrics

Given the 92:8 class distribution, Accuracy is a deceptive metric. We prioritize metrics that emphasize the model's performance on the minority class ($y = 1$) [4, 6]:

- **Area Under the ROC Curve (AUC-ROC):** The primary metric to evaluate the global classification performance.
- **F1-Score:** The harmonic mean of Precision and Recall, defined as:

$$F_1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{6}$$

- **Precision-Recall Trade-off:** Crucial for banking operations where the cost of a False Negative is significantly higher than a False Positive.

5 Experimental Results and Discussion

5.1 Experimental Setup and Parameters

To obtain the results, models were executed through three phases: resampling, hyperparameter tuning, and model evaluation. In the **Resampling Phase**, the specific ratio parameters used were 0.2, 0.3, and 0.5 for SMOTE, Random Under-sampling, and Hybrid SMOTE-Tomek.

In this study, the resampling ratio α is explicitly defined as the target ratio of the number of samples in the minority class (N_{min}) to the number of samples in the majority class (N_{maj}) after resampling is applied:

$$\alpha = \frac{N_{min}}{N_{maj}}$$

For instance, a ratio of $\alpha = 0.5$ dictates that the minority class is resampled (either by synthesizing new data or reducing majority data) until its size equals precisely 50% of the majority class size.

To ensure a fair and robust comparative analysis, a **Hyperparameter Tuning Phase** was integrated prior to final evaluations. We employed `RandomizedSearchCV` to efficiently explore the hyperparameter space for each classifier. Given the highly skewed nature of the data, this search was systematically paired with Stratified K-Fold cross-validation to preserve the original class distribution across all training and validation folds. The hyperparameter optimization objective was set to maximize the ROC-AUC metric.

Finally, for cost-sensitive learning configurations, algorithmic parameters such as `class_weight='balanced'` for Logistic Regression and `scale_pos_weight` for XGBoost and LightGBM were utilized to penalize minority class misclassifications during training.

5.2 Baseline Performance and the Accuracy Paradox

Initially, all algorithms were trained on the original dataset with a severe class imbalance. The results in Table 3 demonstrate the "Accuracy Paradox" [16, 17]: while Accuracy remains high at approximately 92%, the Recall and F1-Scores for the minority class are near zero. For instance, the Standard Random Forest achieved a Validation Accuracy of 91.93% but a Recall of only 0.0004, failing to detect nearly all default cases.

Table 3: Baseline Performance on Validation Set (No Handling)

Model	**Accuracy**	**ROC-AUC**	**Recall**	**F1-Score**
Logistic Regression	0.9196	0.7386	0.0075	0.0147
Random Forest	0.9193	0.7483	0.0004	0.0008
XGBoost	0.9197	0.7599	0.0240	0.0460
LightGBM	0.9199	0.7639	0.0234	0.0450

5.3 Evaluation of Imbalance Handling Strategies

For the linear model (Logistic Regression), SMOTE with a 0.5 ratio improved the F1-Score to 0.2867. In contrast, the Class Weighting strategy for LR yielded a significantly higher Recall of 0.6582. Regarding ensemble models, the Hybrid SMOTE-Tomek approach for XGBoost showed limited effectiveness as the ROC-AUC declined to 0.7174 at a 0.5 ratio. This failure suggests that synthetic noise removal may distort the local data structure for tree-based algorithms. Conversely, Random Under-sampling (RUS) demonstrated superior stability and performance gains across all tree-based architectures.

5.4 Comparative Results and Final Model Selection

Table 4 summarizes the most optimal configurations. Boosting models combined with an Under-sampling ratio of 0.5 provide the highest classification capabilities. **LightGBM with Under-sampling (0.5)** was selected as the final model due to its peak ROC-AUC of 0.7625 and F1-Score of 0.3165.

Table 4: Comparative Results of Best Configurations (Validation Set)

Model	**Strategy**	**ROC-AUC**	**F1-Score**	**Recall**	**Precision**
LightGBM	Under-sampling (0.5)	0.7625	0.3165	0.4425	0.2464
XGBoost	Under-sampling (0.5)	0.7610	0.3115	0.4310	0.2438
Random Forest	Under-sampling (0.5)	0.7467	0.2994	0.3990	0.2397
Logistic Reg.	SMOTE (0.5)	0.7347	0.2867	0.3861	0.2281

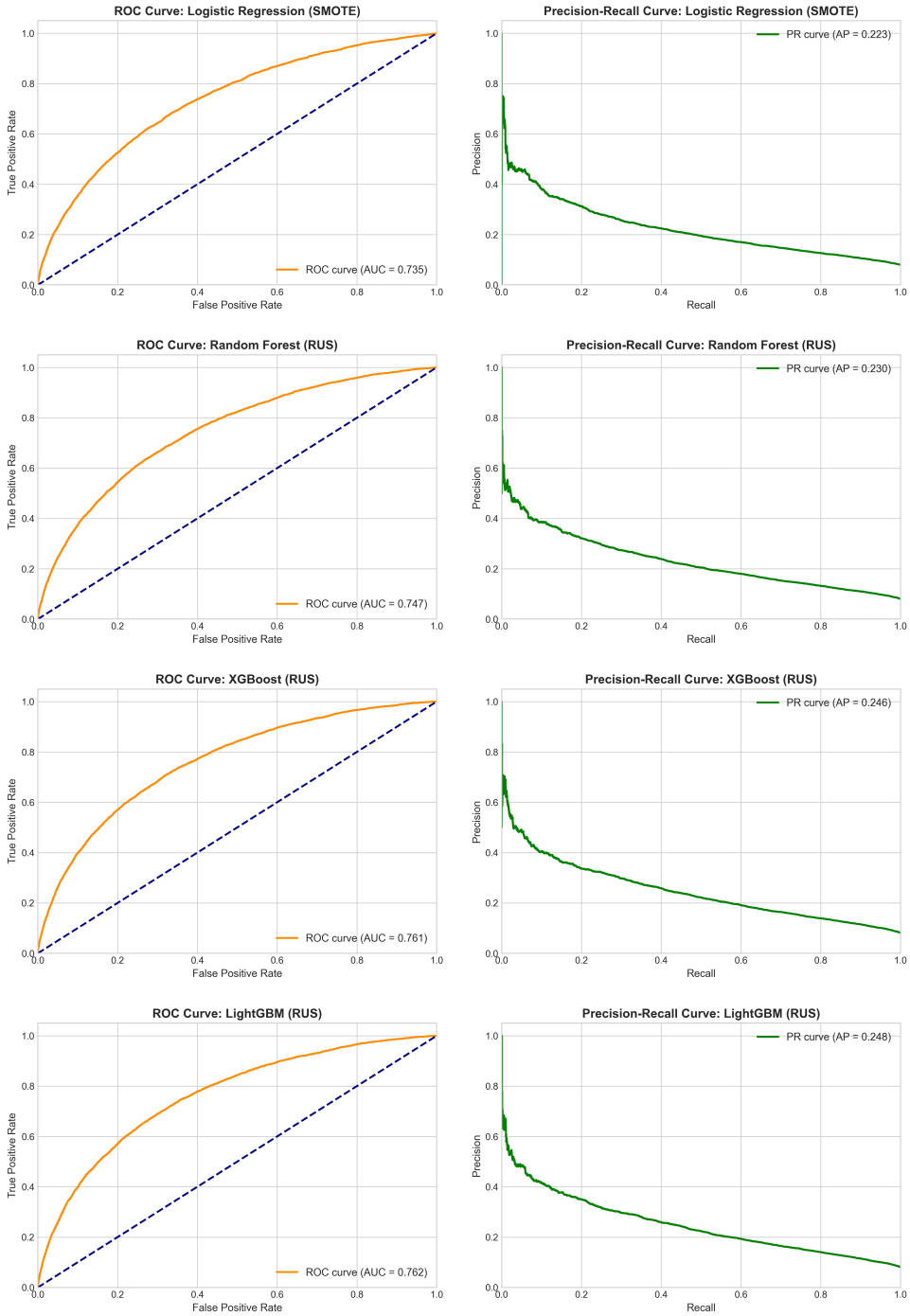

Figure 3: ROC Curves and Precision-Recall Curves Comparison

5.5 Model Interpretability with SHAP

5.5.1 Global Feature Importance and Business Insights

SHAP analysis reveals a clear contrast in the decision-making patterns between linear and tree-based algorithms. For the Logistic Regression model, loan-related characteristics such as `LOG_AMT_CREDIT` (0.694) and `ANNUITY_LENGTH` (0.441) emerge as the primary drivers of default risk. In contrast, advanced ensemble models such as XGBoost and LightGBM rely predominantly on external credit bureau scores to determine borrower risk.

For the best-performing LightGBM model, `EXT_SOURCE_MEAN` emerges as the most influential feature, with a mean absolute SHAP value of 0.4162 under the Under-sampling strategy. SHAP summary plots indicate a clear inverse relationship: higher external credit scores substantially reduce the predicted

probability of default.

Additionally, the presence of `missingindicator_EXT_SOURCE_1` (0.086) among the top-ranked features provides an important financial insight. Borrowers with missing historical credit information, often referred to as "thin-file" customers, tend to exhibit higher predicted risk levels. This result suggests that missing credit history itself serves as a meaningful predictive signal in credit risk assessment.

Table 5: Top Features - SHAP values of the Final LightGBM Model (Under-sampling)

Feature	Ratio 0.2	Ratio 0.3	Ratio 0.5	Mean SHAP
EXT_SOURCE_MEAN	0.4093	0.4163	0.4230	0.4162
ANNUITY_LENGTH	0.1884	0.1928	0.1922	0.1911
EXT_SOURCE_3	0.1594	0.1550	0.1612	0.1585
CREDIT_TO_GOODS_RATIO	0.1234	0.1254	0.1327	0.1272
EXT_SOURCE_1	0.1208	0.1177	0.1224	0.1203
LOG_AMT_GOODS_PRICE	0.1063	0.1072	0.0997	0.1044
EXT_SOURCE_2	0.0993	0.1023	0.1042	0.1020

5.5.2 Feature Importance Shift under Resampling Techniques

Our analysis further shows that synthetic resampling methods can drastically influence the internal decision logic of tree-based models. A prominent anomaly was observed in the XGBoost model: the importance of `CREDIT_TO_GOODS_RATIO` surged from 0.115 in the baseline standard model to a staggering 0.5664 when applying the Hybrid SMOTE-Tomek technique at a 0.5 ratio.

This extreme deviation is not random. It can be explained by the interpolation mechanism of SMOTE operating within specific minority class distributions. If default customers exhibit a highly specific clustering regarding their credit-to-goods ratio (e.g., frequently borrowing amounts that exceed the value of the collateral good), SMOTE will densely interpolate synthetic points precisely around this subspace. Subsequently, Tomek Links clears majority points in this vicinity, creating an artificial, highly separable "island" of default data. Consequently, the greedy splitting criteria of tree-based models over-prioritize this artificially amplified feature, distorting the true financial relationships.

In stark contrast, Under-sampling demonstrates remarkable stability in maintaining feature hierarchy. The patterns observed in the Undersampled LightGBM and XGBoost models strictly parallel those of the standard models, with `EXT_SOURCE_MEAN` consistently retaining its status as the most robust predictor. By merely reducing majority instances rather than synthesizing new ones, Under-sampling preserves the authentic multivariate distribution of the data.

Interestingly, SMOTE exhibited a contrasting, beneficial effect on linear models. For Logistic Regression, the importance of `EXT_SOURCE_MEAN` increased from 0.171 (Standard) to 0.3169 (SMOTE). Because linear models do not rely on local subspace splits but rather on a global hyperplane, injecting synthetic minority instances corrects the model's global perspective, allowing it to better capture the true weight of external credit risk signals.

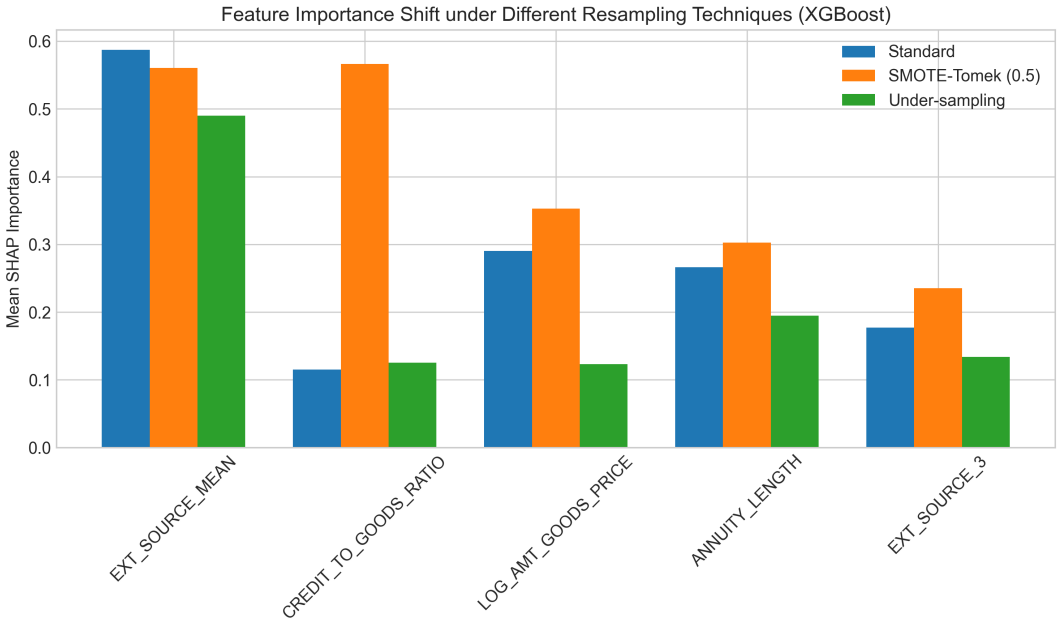

Figure 4: Feature importance shift in XGBoost under different resampling strategies.

5.5.3 Local Interpretability and Decision Instability with LIME

While SHAP provides a comprehensive global view of feature contributions, Local Interpretable Model-agnostic Explanations (LIME) was employed to evaluate the local fidelity of the models. By generating surrogate linear models around specific instances, LIME allows us to investigate how resampling techniques alter the decision-making logic for an individual borrower.

Figure 5 illustrates a striking case study of a single applicant evaluated by the baseline LightGBM model and its resampled counterparts (RUS 0.5 and SMOTE 0.5). The result reveals a severe phenomenon of *decision instability* and *local feature distortion* caused by data balancing techniques.

For the exact same customer profile, the predicted probability of default fluctuates drastically: 21.1%, 66.7% (RUS), and 15.6% (SMOTE). This extreme variance highlights how different resampling mechanisms manipulate the local decision boundaries.

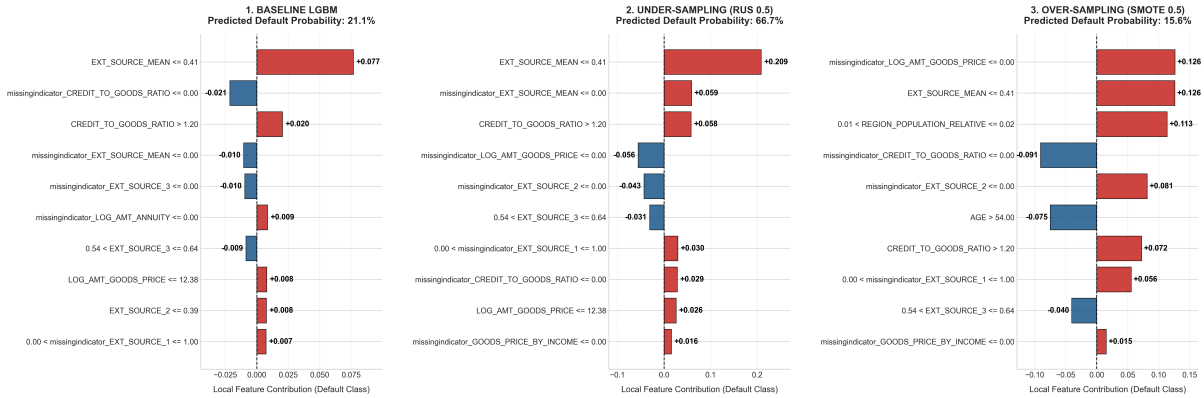

Figure 5: LIME explanations for a single instance across Baseline, Undersampled, and Oversampled LightGBM models. Positive values (red) push the prediction towards default, while negative values (blue) push towards non-default.

A deeper examination of the LIME weights reveals the underlying behavioral shifts of the models:

- **Amplification in Under-sampling:** The RUS 0.5 model does not invent new logic; rather, it aggressively amplifies existing risk signals. The penalty weight for `EXT_SOURCE_MEAN` surged from

+0.077 in the baseline to +0.209. By removing a massive portion of the "safe" majority class, the model becomes hyper-sensitive to marginal risk indicators.

- **Logic Scrambling in Over-sampling:** SMOTE 0.5 completely alters the local narrative. While `EXT_SOURCE_MEAN` remains significant, the model abruptly prioritizes entirely different features, such as `missingindicator_LOG_AMT_GOODS_PRICE` (+0.126) and demographic variables like `REGION_POPULATION_RELATIVI` (+0.113), which were largely ignored by the baseline model. This suggests that SMOTE's synthetic generation creates artificial local neighborhoods, forcing the tree algorithm to split on secondary features that may lack genuine financial causality for this specific customer.

From a managerial and regulatory perspective, these findings are critical. If a financial institution relies on an oversampled model and uses XAI to explain a loan rejection to a customer, the generated explanation may be mathematically accurate for the synthetic space, but fundamentally misaligned with the true market reality. This trade-off between predictive recall and local explainability stability must be carefully weighed in automated credit scoring systems.

6 Conclusion

This study comprehensively evaluated the dual impact of various imbalance handling techniques on predictive performance and model explainability within the domain of credit scoring. Traditional approaches often focus solely on maximizing predictive metrics, neglecting the "Accuracy Paradox" and the transparency required in financial contexts.

Regarding predictive performance, our empirical results established that the combination of Gradient Boosting architectures and Random Under-sampling achieves the optimal equilibrium. Specifically, LightGBM coupled with an under-sampling ratio of 0.5 emerged as the superior model, yielding a ROC-AUC of 0.7625 and effectively navigating the essential trade-off between Recall and Precision.

Crucially, from the perspective of Explainable AI (XAI), this research highlights that resampling strategies are not merely data pre-processing steps; they act as profound modifiers of algorithmic behavior. We discovered that synthetic data generation methods, such as SMOTE and SMOTE-Tomek, introduce severe "Feature Importance Shifts" in tree-based models. By artificially amplifying specific sub-spaces (such as the `CREDIT_TO_GOODS_RATIO`), these techniques force models to overfit on synthetic noise, thereby degrading the reliability of the model's financial logic. Conversely, Under-sampling proved to be exceptionally stable, preserving the authentic hierarchy of predictive features (e.g., maintaining `EXT_SOURCE_MEAN` as the primary indicator) without distorting the underlying data distribution.

Ultimately, this study concludes that for high-stakes financial applications, the synergy of Under-sampling and LightGBM not only delivers robust default detection but also guarantees the structural transparency and interpretability essential for regulatory compliance.

While this research provides valuable insights, it is inherently limited to a single institutional dataset. Future work should validate these findings across diverse financial datasets to ensure broader generalizability. Additionally, subsequent research could focus on quantitatively measuring the localized instability of LIME explanations across the entire minority class using robust stability metrics. Finally, exploring advanced generative models, such as Tabular GANs, could offer a modern comparative baseline against traditional SMOTE techniques regarding their impact on explainability.

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